Project Two ReadMe
CS-340 SNHU
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About the Project/Project Title

This project creates an interactive data dashboard for Grazioso Salvare using Python, Dash, and MongoDB. The dashboard connects to the Animal Adoption Center (AAC) database through a reusable CRUD Python module (AnimalShelter). The dashboard allows users to filter rescue-ready dogs, view updated data tables, and visualize results through dynamic charts and a geolocation map.

This dashboard builds on the CRUD module developed in Project One.

Motivation

The purpose of this project is to provide Grazioso Salvare with a functional dashboard that supports the identification of dogs that are suitable for rescue training. The dashboard enables real-time filtering of animals based on rescue type, dynamic updates to charts and maps, and a clean, user-friendly interface for decision-making. Project two demonstrates that data visualizations, CRUD operations, and interactive UI components can work together for real-world needs.

Getting Started

To run this dashboard, the user must have access to a MongoDB instance containing the AAC dataset imported from the aac_shelter_outcomes.csv file. The MongoDB account created in Module Three is hardcoded and used to authenticate the connection, and the AnimalShelter class handles all database interactions. The dashboard is built using Dash and JupyterDash, which allow it to run directly inside a Jupyter Notebook. The Codio environment already includes most of the required tools, so setup is minimal. Once the CRUD module is imported and instantiated, the dashboard loads the initial dataset and becomes fully interactive.

Installation

The Codio environment for CS-340 includes all necessary tools for running this project, so installation requirements are minimal. Python 3 is used for the CRUD module and dashboard, and PyMongo enables communication between Python and MongoDB. Dash, Plotly, and Dash Leaflet are used for building the interactive interface, generating charts, and rendering the geolocation map. With these tools in place, the dashboard can be executed directly from the Jupyter Notebook without additional setup.

Usage

The dashboard begins by importing and instantiating the AnimalShelter class, which establishes a secure connection to the AAC database. The initial dataset is loaded using the read() method and converted into a Pandas DataFrame. The user interacts with the dashboard through a set of radio buttons that represent the available rescue filters. When a filter is selected, a callback function constructs the appropriate MongoDB query and retrieves the filtered results. These results automatically update the data table, which in turn updates the pie chart showing breed distribution and the geolocation map showing the selected animal's coordinates. The entire dashboard responds dynamically to user input, ensuring that all visualizations remain synchronized with the underlying data.

Tests

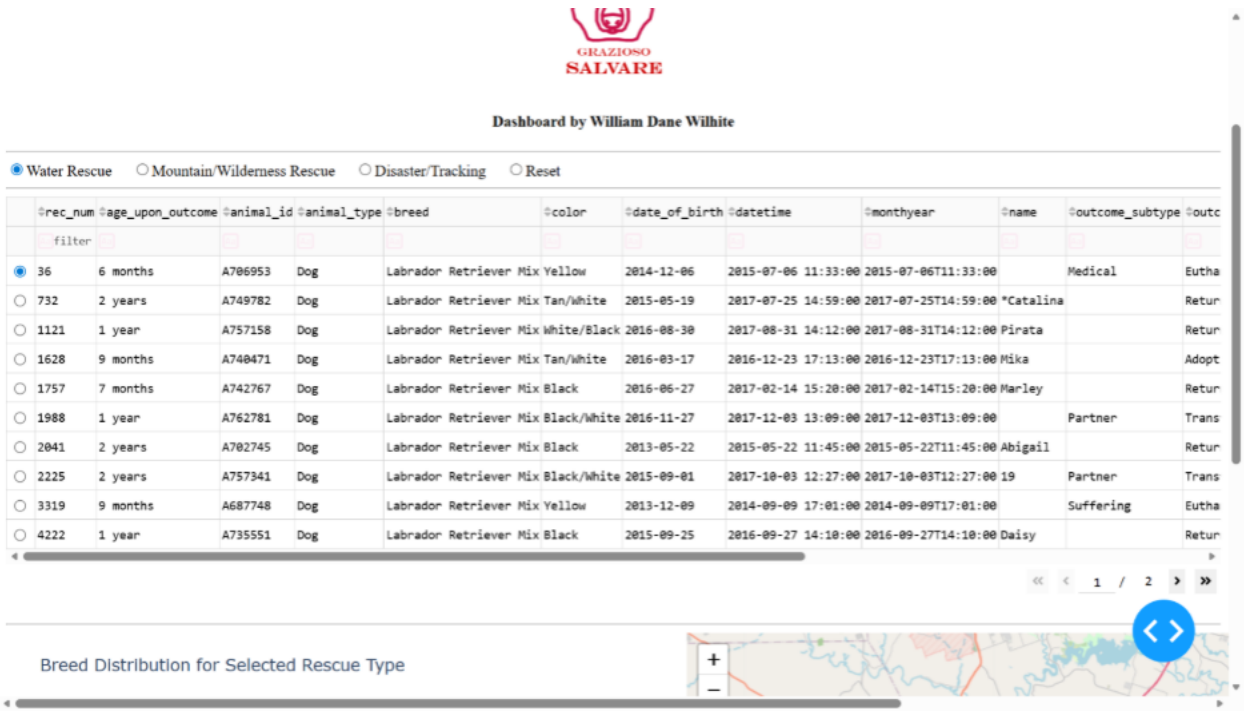
Some testing was performed in the ModuleFourTestScript.ipynb notebook. The notebook instantiated. Project two was tessted by running the ProjectTwoDashboard.ipynb after corrections and completion to test for functionality. Once functional, screenshots were taken to prove adequacy.

Code Example

The core logic of the dashboard relies on a callback that updates the DataTable based on the selected rescue filter. This callback uses the build_query() function to generate the correct MongoDB query and then calls the read() method to retrieve matching records. The results are cleaned by removing the MongoDB _id field, which prevents rendering issues in Dash. Additional callbacks update the pie chart and map whenever the DataTable changes. These components work together to create a seamless and interactive experience for the user, ensuring that all dashboard elements remain consistent with the selected filter.

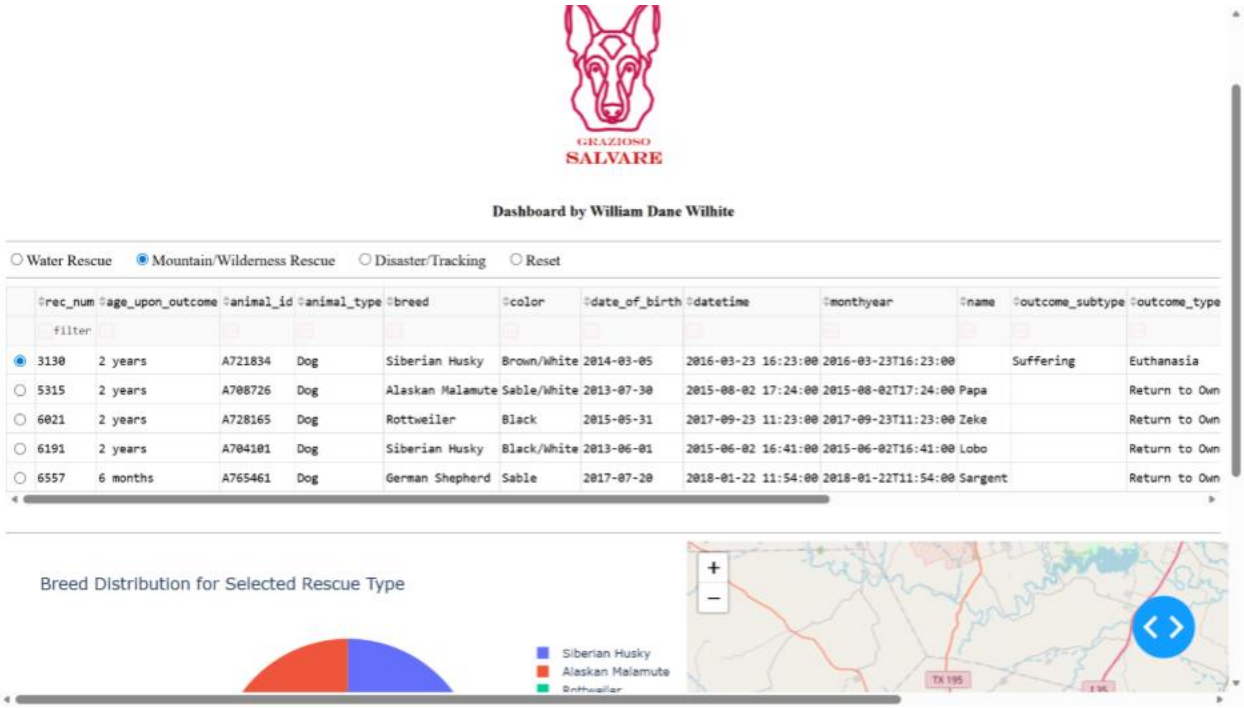
Screenshots

Water Rescue:

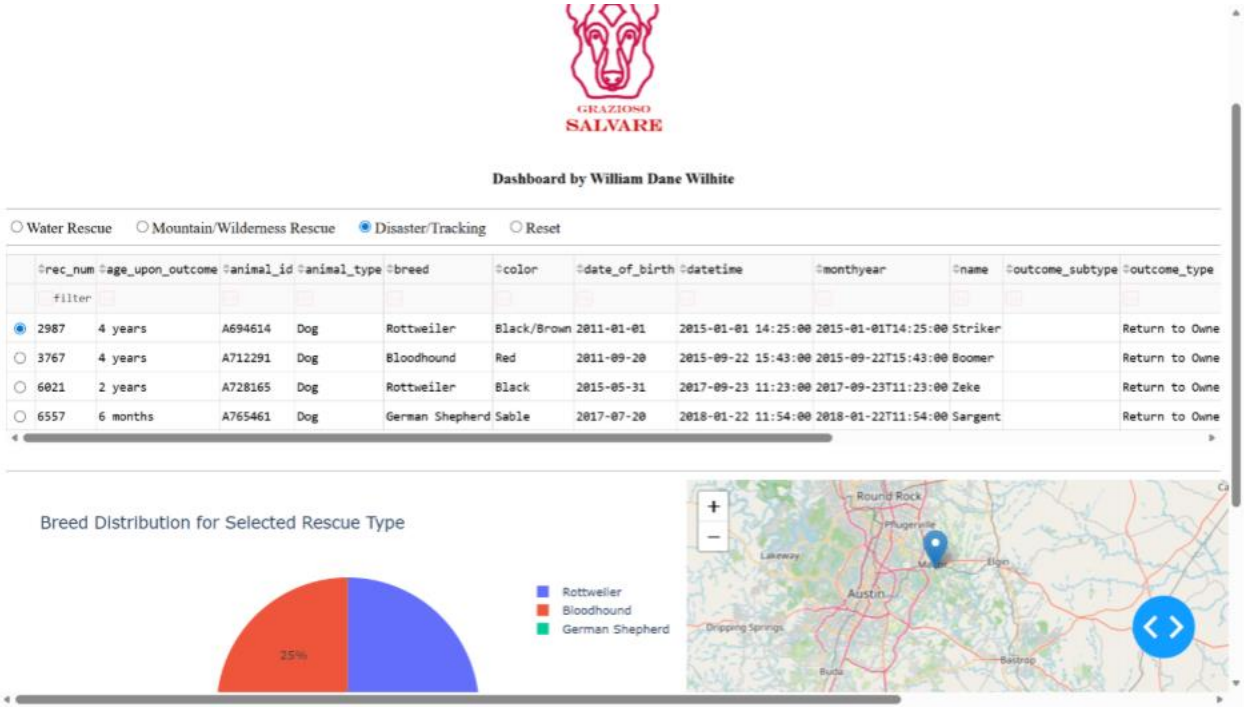


rec_num	age_upon_outcome	animal_id	animal_type	breed	color	date_of_birth	datetime	monthyear	name	outcome_subtype	outc
filter											
36	6 months	A706953	Dog	Labrador Retriever Mix	Yellow	2014-12-06	2015-07-06 11:33:00	2015-07-06T11:33:00		Medical	Eutha
732	2 years	A749782	Dog	Labrador Retriever Mix	Tan/White	2015-05-19	2017-07-25 14:59:00	2017-07-25T14:59:00	*Catalina		Retur
1121	1 year	A757158	Dog	Labrador Retriever Mix	White/Black	2016-08-30	2017-08-31 14:12:00	2017-08-31T14:12:00	Pirata		Retur
1628	9 months	A740471	Dog	Labrador Retriever Mix	Tan/White	2016-03-17	2016-12-23 17:13:00	2016-12-23T17:13:00	Mika		Adopt
1757	7 months	A742767	Dog	Labrador Retriever Mix	Black	2016-06-27	2017-02-14 15:20:00	2017-02-14T15:20:00	Marley		Retur
1988	1 year	A762781	Dog	Labrador Retriever Mix	Black/White	2016-11-27	2017-12-03 13:09:00	2017-12-03T13:09:00		Partner	Trans
2041	2 years	A702745	Dog	Labrador Retriever Mix	Black	2013-05-22	2015-05-22 11:45:00	2015-05-22T11:45:00	Abigail		Retur
2225	2 years	A757341	Dog	Labrador Retriever Mix	Black/White	2015-09-01	2017-10-03 12:27:00	2017-10-03T12:27:00	19	Partner	Trans
3319	9 months	A687748	Dog	Labrador Retriever Mix	Yellow	2013-12-09	2014-09-09 17:01:00	2014-09-09T17:01:00		Suffering	Eutha
4222	1 year	A735551	Dog	Labrador Retriever Mix	Black	2015-09-25	2016-09-27 14:10:00	2016-09-27T14:10:00	Daisy		Retur

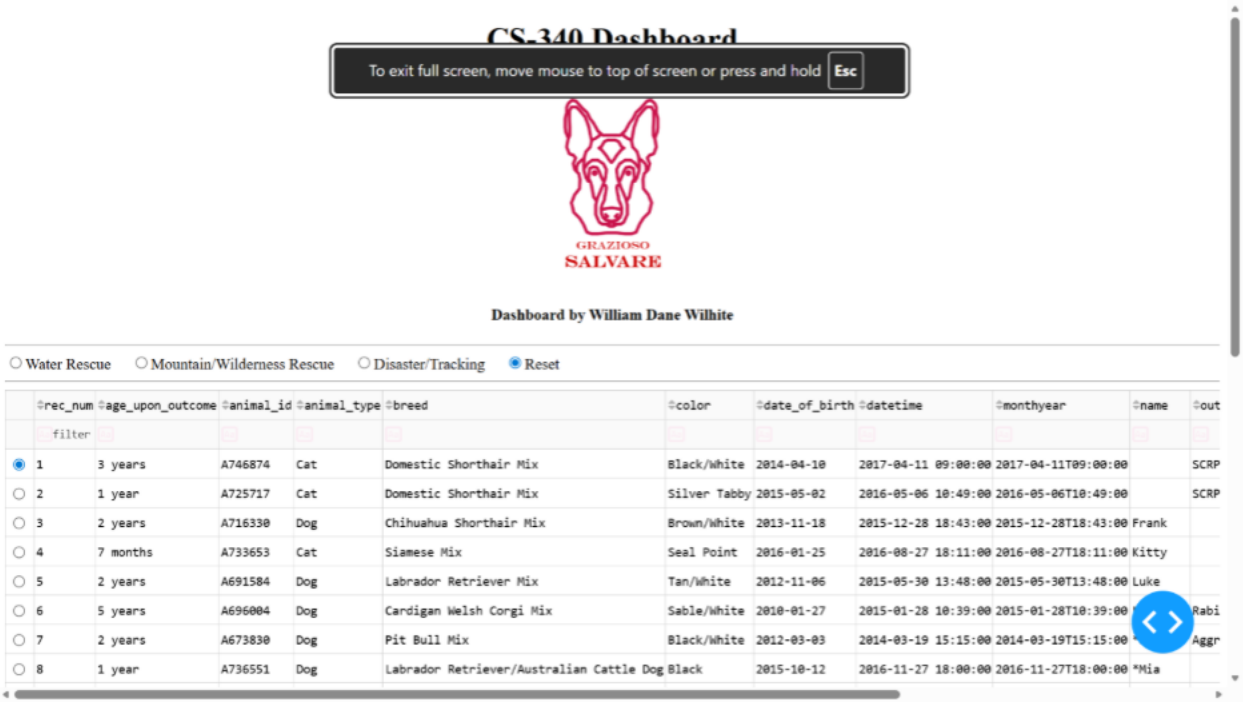
Mountain Wilderness Rescue:



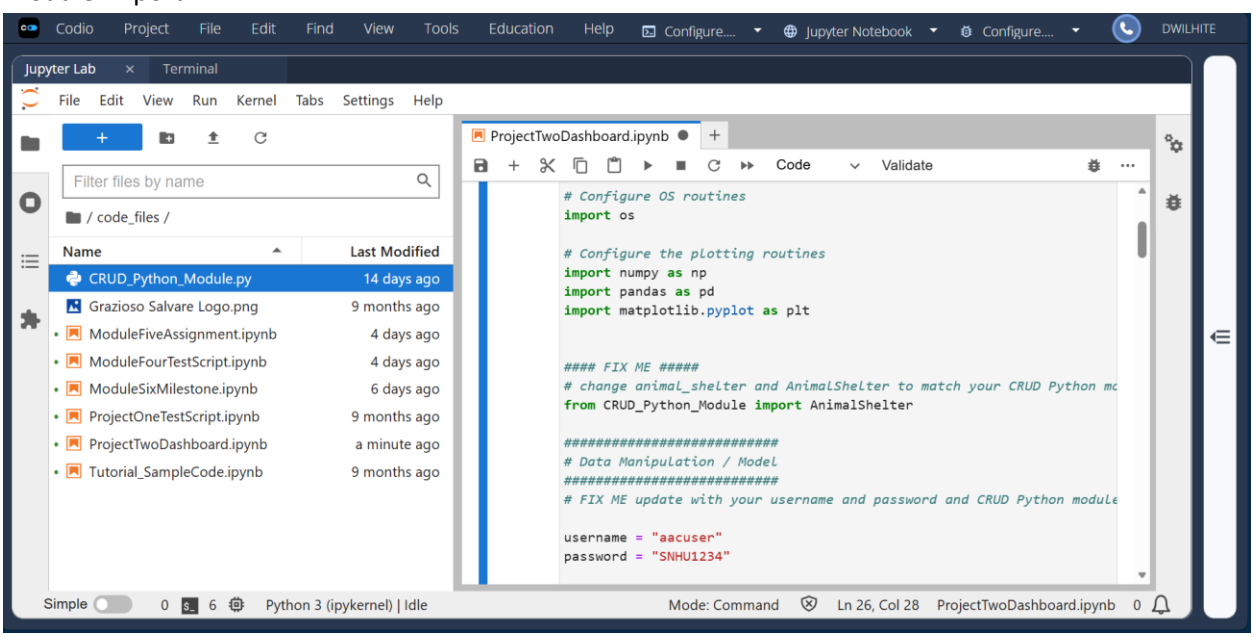
Disaster Tracking:



Reset:



Module Import:



Contact:

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